

Gratitude Growth: The Impact of Gratitude Journaling on High School Freshman Happiness

DATASCI 241

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Abstract

Studies have shown that people tend to focus more on hardships and problems rather than the positives in daily life. This effect is further perpetuated in teenagers who often rely on social media for connectedness and relationships among friends. Gratitude journaling has previously been shown to improve life satisfaction, decrease stress, and improve social connectedness by increasing resilience to negative emotions. This randomized controlled trial aimed to investigate the effects of gratitude journaling on gratitude and happiness in a cohort of 107 high school freshmen. Students were blocked by gender, class section, and class grade. Both treatment and control groups journaled the first 5 minutes of class, with the control receiving academic related/neutral prompts and the treatment receiving happiness/gratitude prompts. Each student completed four journal entries per week over the course of a four week trial. Contrary to expectations, our analysis revealed no significant difference in happiness levels between the treatment and control groups at the conclusion of the intervention compared to the baseline. This finding highlights the complexity of influencing gratitude through structured school based interventions.

Objective/ Purpose

This study aimed to investigate the impact of gratitude journaling on the happiness levels of high school freshmen. Building upon previous research indicating the efficacy of gratitude practices in enhancing life satisfaction, reducing stress, and improving social connectedness (Miller)(Emmons & McCullough), this randomized controlled trial sought to understand whether gratitude journaling can similarly influence the happiness of teenagers. Our primary research question was as follows: *Does daily gratitude journaling lead to increased happiness levels among high school freshmen?* We hypothesized that participants assigned to the treatment group who engaged in gratitude journaling would experience a greater increase in happiness levels compared to those in the control group, who journaled about neutral academic topics. This hypothesis is grounded in the premise that focusing on gratitude fosters a positive mindset and cultivates resilience to negative emotions, ultimately leading to greater subjective well-being.

Experiment Design

To assess our research question and evaluate our hypothesis, a randomized controlled trial was used to study the impact of gratitude journaling on happiness. Happiness was operationalized as

a metric using the Oxford Happiness Questionnaire, which ranges from 1 to 6, with 6 being the highest possible happiness score (1 “strongly disagrees” to 6 “strongly agree”). Three additional gratitude questions were added to the Oxford Happiness Questionnaire to assess students' gratefulness. The additional gratitude questions ranged from 1 to 6, 6 being the most grateful score. Appendix 2 and Appendix 3 highlight the questions within the questionnaires. The final questionnaire had an additional five questions aimed at student perception of effort and their own self-evaluation of the effect of the treatment, journaling on their academics, and emotions.

Journals were purchased for each student with their names labeled on the outside of the journal. The journals were kept in the classroom at all times with new prompts taped in every morning before students arrived at school, with the treatment group getting a prompt about gratitude and the control group getting a prompt about academics. Students were unaware that they were divided in these groups and were being given differing prompts. At the start of each class, students were asked to journal for 5 minutes, allowing them time to read the prompt, think about it, and then write. Students' responses ranged from a few sentences to full-page journaling. In the classroom, students were divided by treatment and control groups, seating treatment group students at a lab table together, and control group students on other lab tables together. This was done in an attempt to reduce spillover if students looked at or discussed each other's journal prompts. Student privacy was protected, and journal entries were not noted nor recorded. Additionally, students were informed that this exercise would not be graded and they could retain their journals after the study concluded.

The treatment group had specific gratitude prompts, allowing them to think deeply about people, items, events, situations, and other aspects of their lives that bring them gratitude. This aimed to foster a gratitude mindset and reflect on positive experiences or feelings. The control group was given a prompt to write about school or academic-related topics including setting goals and evaluating their set goals. This method ensured that both groups were engaged in the act of writing, but the control content focuses on neutral or non-emotionally charged topics rather than gratitude. The core difference between the treatment and control groups lies in the content of the journaling prompts. Table 1 of the appendix highlights the exact journaling prompts used for both treatment and control groups, along with timelines of measurement. All participants journaled for 5 minutes, 4 days a week, over the course of 4 weeks with a two week off period due to spring break.

Comparison of Potential Outcomes

We compared the changes in happiness levels, as measured by the Oxford Happiness Questionnaire with additional gratitude questions, between the treatment group and the control group. Specifically, we examined whether there was a significant difference in the mean change in happiness scores from baseline to the conclusion of the intervention between the treatment and control groups.

The ROXO comparison then looks like the following:

R	X	O
Treatment Group	Gratitude Journaling	Happiness Outcomes
R	X	O
Control Group	Neutral Journaling	Happiness Outcomes

Recruitment

107 participants, 55 females and 52 males, ranging in age from 14 to 15. The demographics of the groups are as follows: 84% identify as Hispanic (90/107), 12% identify as Black Americans (13/107), 2% identify as Asian Indian (2/107), and 2% identify as Albanian (2/107). These individuals are part of a high school, Cristo Rey New York High School, and distributed amongst their four biology sections. The sections are grouped based on math proficiency on standardized tests, map scores, and math taken in their elementary schools. All participants are from low socioeconomic income homes, of families of four, and annual income of \$51,338 or less. Participants come from a range of areas in New York City, predominantly Harlem/ Manhattan 53% (57/107) and the Bronx 37% (40/107).

Randomization

The entire sample of students was used in the randomization process, which began by blocking on the following covariates: class section, gender, race, and current letter grade in biology. After blocking and random assignment (done using random sampling in Python with numpy package), 58 students were assigned to treatment and 49 to control (n=107), Table 1.

Class Section	Treatment	Control
R	15	13
I	14	12
N	15	12
W	14	12

Table 1: Distribution of treatment and control groups per class section

There was a slight increase in individuals in treatment due to uneven balances among other blocking metrics including gender and race, thus this numerical assignment provided the most balance.

Blocking the class section was an important part of the randomization process in order to ensure clustering effects were mitigated. Additional covariate blocking helped reduce biases that could occur in our result if inherent differences exist for gratitude based on gender, race, or academic level. Participants remained in their assigned group for the duration of the experiment.

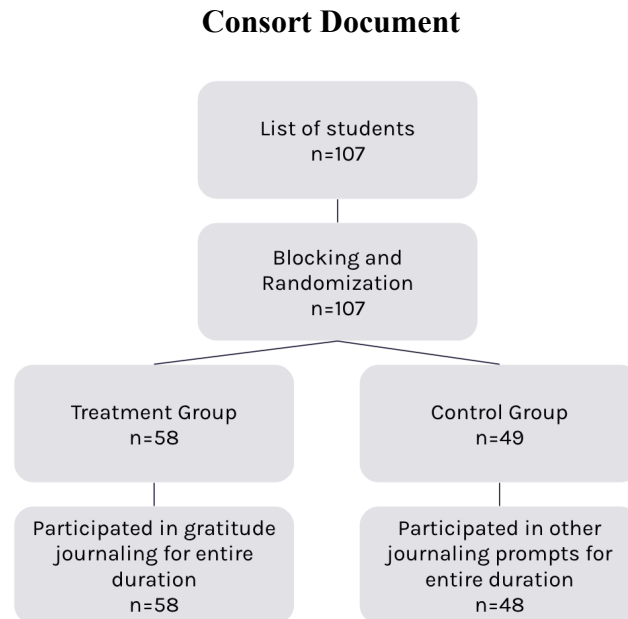


Figure 1: Outline of sampling and assignment for treatment and control

The consort document outlines the sampling and assignment procedure, Figure 1. Note that one student in the control group moved away two weeks after the treatment started. No further data was collected, and this student was removed from all subsequent analyses.

Power Calculations

Before beginning our experimentation and analysis, power calculations were conducted to formulate some assumptions about effect size and the anticipated experimental power. The three effect sizes we chose for our analysis included 0.2, 0.5, and 0.8. These sizes were chosen based on previously published research, specifically the article, “Using Effect Size—or Why the P Value Is Not Enough” (Sullivan & Feinn). The article discusses common effect size indices specifying Cohen’s term d and the curation of small/medium/large magnitude effect sizes to formulate discernible differences in treatment effect - power of the study. Using the three effect sizes, we simulated baseline data using the three covariate blocking implemented at the start of the study (gender, school grade, and class section). As well as incorporating our happiness/gratitude scores with continuous value 0 to 6. Sample size for the simulation, when calculating power, ranged from 30 to 100 by increment of 10 to observe the full range. In order to simulate results, our test of choice was a regression model with post-treatment happiness scores as the dependent variable and condition as the independent variable, along with covariates

of pre-treatment happiness scores (baseline), grade, gender, and class section. These covariates were included in an attempt to explain more of the variability in our data. After running our simulations, we find that based on our constant sample size, a small effect size has less than 10% statistical power, a medium effect size has around 25% statistical power, and a large effect size has around 55% statistical power, Figure 2. In order to find significant results in our experimentation, a larger treatment effect will need to be observed. Given the two week timeline of our study, and based on prior gratitude journaling research, an effect size as large as 0.8 may not be viable to observe.

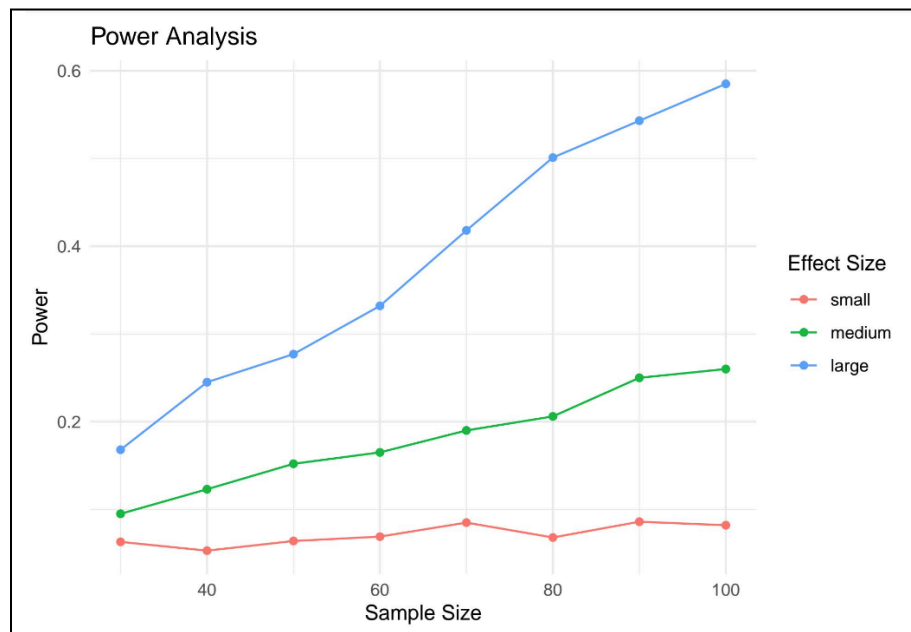


Figure 2: Potential power of experiment varied by potential sample sizes distributed by effect size

Results

Data

Baseline student data collection involved the curation of student demographics, as mentioned above, for the use of blocking. Baseline happiness scores were taken prior to beginning experimentation. After every week of journaling, happiness, and gratitude scores were taken again to measure any progress or changes from treatment. Final happiness scores were collected at the end of the 4 weeks (14 class meetings). In addition to the happiness scores, additional questions were added to the survey focused on assessing any changes in the feeling of gratitude. Appendix 2 and Appendix 3 state the questions in the Oxford Happiness Survey, the additional gratitude statements, and the final self-evaluation questions. The following instructions were given to students,

“Instructions.

Below are a number of statements about happiness. Please indicate how much you agree or disagree with each by entering a number in the blank after each statement, according to the following scale:

1 = strongly disagree 2 = moderately disagree 3 = slightly disagree 4 = slightly agree 5 = moderately agree 6 = strongly agree

Please read the statements carefully, some of the questions are phrased positively and others negatively. Don't take too long over individual questions; there are no “right” or “wrong” answers (and no trick questions). The first answer that comes into your head is probably the right one for you. If you find some of the questions difficult, please give the answer that is true for you in general or for most of the time.” (Hills, P., & Argyle, M.)”

Scoring was conducted based on the instructions from the Oxford Happiness Survey. Details of scoring are included in Figure 2 of the appendix. At the final survey collection, in addition to happiness and gratitude questions, additional metrics were collected on student wellbeing and effort. Surveys were answered by students after a 5-minute journaling time on the last day of the week before class materials were undertaken. Surveys were converted into a Google form in order to be distributed to the students through their email. Data was collected on Google Sheets and processed, the scoring process is explained in Figure 2 in the appendix. The scores are then converted into csv and uploaded for analysis. Students used computers, laptops, tablets, and cell phones to complete the surveys. If a student was absent during class time they would complete the survey during the next class.

Analysis

OLS regressions with robust standard errors were used to evaluate the impact of gratitude journaling on happiness creating increasingly complex models which control for baseline happiness scores, covariates (location, ethnicity, gender, grade), and effort. This regression measures the impact of being assigned to the treatment condition (gratitude journaling) on final happiness scores at the end of the 4 week period.

From the first stargazer, Figure 3, we find that the treatment condition of gratitude journaling was not statistically significant across all four regressions for estimating happiness, though the coefficient remained consistent across all models. Baseline happiness scores remained the most consistent and highly statistically significant (at the 1% level) across all models, implying that prior happiness scores were the strongest predictor of final happiness scores. We also saw a weaker statistical significance (at the 10% level) of students who are Hispanic estimating a slight decline among happiness scores. This is a weaker significance and was not found among other ethnicities. We do not attribute this significance to any justifiable difference.

	Dependent variable:			
	Happiness Score			
	(1)	(2)	(3)	(4)
Condition	0.076 (0.249)	-0.011 (0.132)	0.087 (0.139)	0.078 (0.139)
Preferred_Gender_Male			-0.178 (0.139)	-0.162 (0.140)
Location_Manhattan			0.152 (0.155)	0.124 (0.158)
Location_Queens			0.242 (0.259)	0.220 (0.260)
Ethnicity_Asian_Indian			-1.060 (0.722)	-1.030 (0.722)
Ethnicity_Black_American			-0.631 (0.532)	-0.534 (0.541)
Ethnicity_Hispanic			-0.963* (0.508)	-0.896* (0.512)
Class.Section_N			-0.091 (0.200)	-0.087 (0.199)
Class.Section_R			0.085 (0.193)	0.103 (0.193)
Class.Section_W			0.272 (0.220)	0.297 (0.221)
letter_grade_B			0.294 (0.284)	0.316 (0.285)
letter_grade_C			0.339 (0.299)	0.362 (0.300)
letter_grade_D			0.474 (0.405)	0.478 (0.405)
b1.happ		0.901*** (0.108)	0.904*** (0.115)	0.877*** (0.118)
effort				0.085 (0.082)
Constant	3.940*** (0.198)	0.457 (0.430)	0.918 (0.742)	0.649 (0.787)
Observations	106	106	106	106
R2	0.002	0.402	0.468	0.474
Adjusted R2	-0.008	0.391	0.386	0.387
Residual Std. Error	0.867 (df = 104)	0.674 (df = 103)	0.677 (df = 91)	0.676 (df = 90)
F Statistic	0.201 (df = 1; 104)	34.700*** (df = 2; 103)	5.720*** (df = 14; 91)	5.420*** (df = 15; 90)
Note:	*p<0.1; **p<0.05; ***p<0.01			

Figure 3: Stargazer - Regressions on happiness score (note that b1.happ refers to baseline happiness scores)

Next, we ran an OLS regression with robust standard errors to evaluate the impact of gratitude journaling on feelings of gratitude specifically. We chose to do this to assess whether gratitude journaling has some inherent impact on general feelings of gratitude among students that were unexplained or different from feelings of happiness. Following the same procedure as above, this regression measures the impact of being assigned to the treatment condition (gratitude journaling) on final gratitude scores (only gratitude questions) at the end of the 4 week period with increasingly complex models.

	Dependent variable:			
	Gratitude Score			
	(1)	(2)	(3)	(4)
Condition	0.222 (0.249)	0.021 (0.208)	0.095 (0.217)	0.089 (0.218)
Preferred_Gender_Male			-0.353 (0.217)	-0.337 (0.220)
Location_Manhattan			0.575** (0.241)	0.550** (0.245)
Location_Queens			0.530 (0.400)	0.499 (0.404)
Ethnicity_Asian_Indian			-1.770 (1.120)	-1.740 (1.120)
Ethnicity_Black_American			-1.180 (0.827)	-1.080 (0.843)
Ethnicity_Hispanic			-1.200 (0.789)	-1.140 (0.798)
Class.Section_N			-0.214 (0.309)	-0.206 (0.310)
Class.Section_R			-0.300 (0.299)	-0.285 (0.301)
Class.Section_W			0.191 (0.339)	0.207 (0.341)
letter_grade_B			0.147 (0.441)	0.163 (0.443)
letter_grade_C			-0.127 (0.463)	-0.113 (0.465)
letter_grade_D			-0.100 (0.631)	-0.108 (0.633)
b1.grad		0.665*** (0.104)	0.685*** (0.109)	0.671*** (0.112)
effort				0.079 (0.127)
Constant	4.550*** (0.198)	1.480*** (0.503)	2.440** (1.090)	2.170* (1.180)
Observations	106	106	106	106
R2	0.008	0.290	0.378	0.381
Adjusted R2	-0.002	0.277	0.283	0.278
Residual Std. Error	1.240 (df = 104)	1.050 (df = 103)	1.050 (df = 91)	1.050 (df = 90)
F Statistic	0.840 (df = 1; 104)	21.100*** (df = 2; 103)	3.960*** (df = 14; 91)	3.690*** (df = 15; 90)
Note:				*p<0.1; **p<0.05; ***p<0.01

Figure 4: Stargazer - Regressions on gratitude score (note that b1.grad refers to baseline gratitude scores)

From the second stargazer, Figure 4, we find that the treatment condition of gratitude journaling was not statistically significant across all four regressions, and its impact decreased with the addition of controls. Baseline gratitude scores remained the most consistent and highly statistically significant (at the 1% level) across all models, implying that prior happiness scores were the strongest predictor of final happiness scores. We were additionally surprised to find that residing in Manhattan (LocationM) was significant at the 5% level for predicting final gratitude scores. We are unsure why this variable resulted in significance with gratitude, though we believe it could perhaps be a spurious correlation.

After noticing some spurious correlations with covariates in the regression as aforementioned which were blocked prior to treatment assignment, we ran a covariate balance check to evaluate

the balance of covariates among treatment and control groups and assess for potential sources of bias or confounders. The plot assessing covariate balance, Figure 5, indicates that the means for most covariates are comparable between the treatment and control groups, with standardized mean differences close to zero for many variables. However, there are some discrepancies in variables such as ethnicity, suggesting potential imbalances in these factors between the groups. We don't believe this is significant enough to lead to major biases given the mean differences are under ± 0.2 .

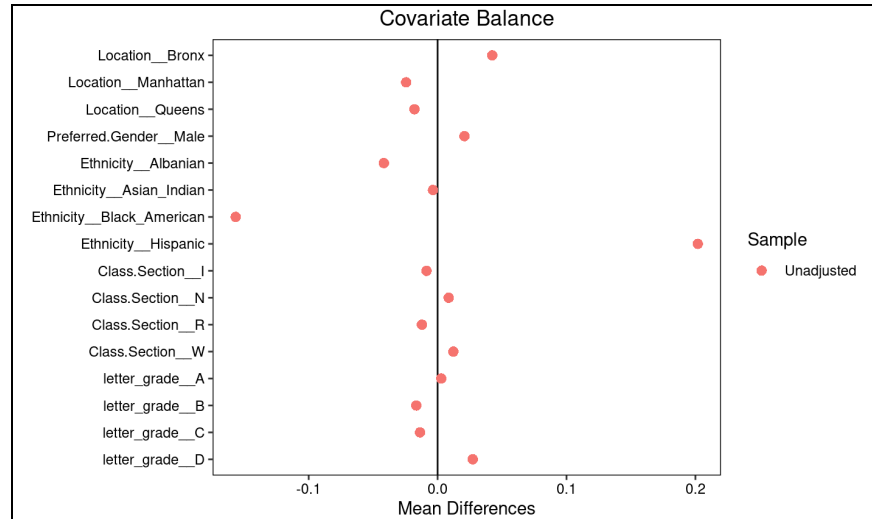


Figure 5. Covariate balance among all potential controls

Next, we moved on to analyze time based impacts on happiness scores over the course of the 4 weeks of treatment. Having collected weekly student surveys, we were able to add in time as a variable of measure by weeks, or the time at which each survey was taken. Figure 6 shows happiness scores box plots from baseline to final measurements for treatment and control groups. The scores seem similar overall and range in variance, but we do notice a slight visual increase in scores at measurement 4 and 5 for the treatment group. The line plot below provides a better demonstration of the slight differences in happiness score by treatment group assignment. Though note that the y-axis demonstrates these changes are small.

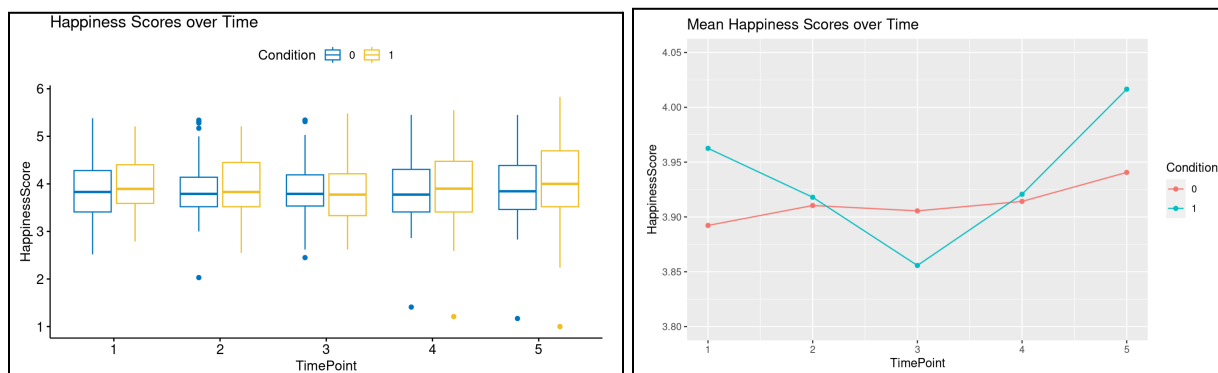


Figure 6. Distribution of happiness scores over time and mean happiness scores over time

Note that we see a dip among happiness scores for the gratitude journaling group whereas the control group sees a slight consistent increase among timepoints (measurement points). We additionally noticed the difference among happiness scores at baseline (time point 1) and conducted a simple T test to assess for statistical significance. From the results of the t-test, Figure 7, we find that the difference in means for happiness scores at baseline among treatment and control is not statistically significant, which helps confirm that our group assignment is done well and not attributable to happiness score differences over time.

Welch Two Sample t-test	
data: b1.happ by Condition	
t = -0.8, df = 97, p-value = 0.4	
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0	
95 percent confidence interval:	
-0.335 0.141	
sample estimates:	
mean in group 0	mean in group 1
3.87	3.96

Figure 7. Welch Two Sample t-test for Differences in Baseline Happiness Scores For Treatment and Control

To test for whether these visual observations are indeed of significance, we performed a repeated measures analysis to study the impact of happiness score changes over time and with condition as an interaction. A two way repeated measures ANOVA test was used to study the score changes over time. Our regression measured the impact of treatment assignment, time point, and the interaction among treatment assignment and time point on final happiness scores. The summary table below, Figure 8, demonstrates that time alone and the interaction between time and treatment assignment are both not statistically significant.

Error: Student.ID					
	Df	Sum Sq	Mean Sq		
Condition	1	2.1	2.1		
Error: Within					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Condition	1	0.1	0.060	0.11	0.74
TimePoint	1	0.1	0.122	0.23	0.63
Condition:TimePoint	1	0.0	0.000	0.00	0.99
Residuals	524	277.7	0.530		

Figure 8. Two way repeated measures ANOVA test to identify potential changes in happiness scores over time

At final measurement students were also asked to rate their effort on journaling and whether or not they experienced increased well being after journaling for the past 4 weeks. The box plot below demonstrates average effort and well being scores for both treatment and control. Here,

we notice that even though both groups state similar effort exertion among the journaling exercise, the treatment group which participated in gratitude journaling saw an almost 1 unit increase in wellbeing compared to the control group.

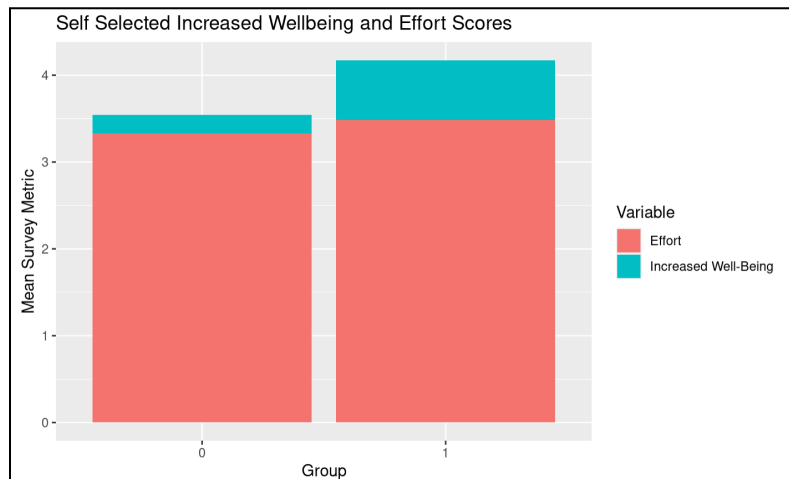


Figure 9. Bar plot of effort and increases well being scores among treatment (1) and control (0) groups

To operationalize this further, formal t-tests were conducted for comparison. Effort was not statistically significant between treatment and control. However, to our surprise, we find that increased well being did indeed have statistically significant differences among treatment and control at the 5% significance level. With a p-value of 0.02 we see that the group that participated in gratitude journaling had a mean increased well being metric of 4.17 as compared to 3.54 for the control group.

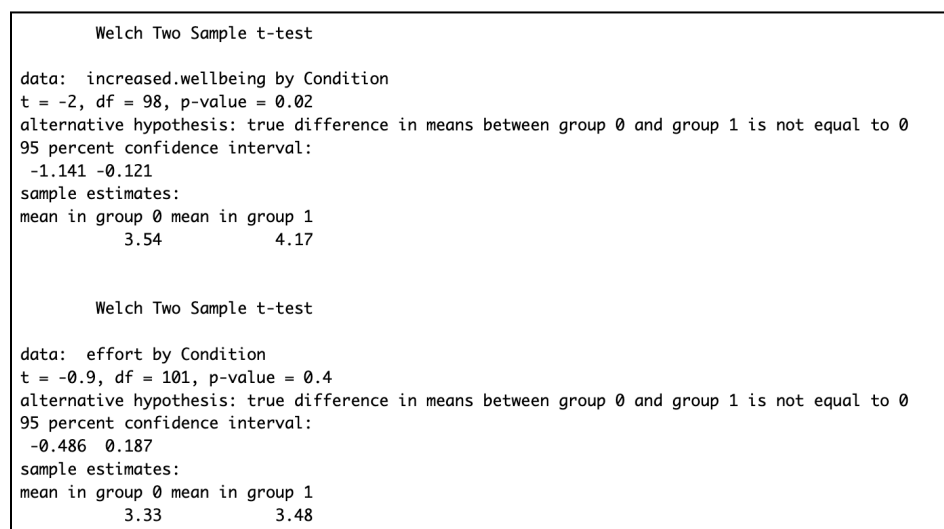


Figure 10. Welch Two Sample t-tests for differences in effort and wellbeing among treatment and control groups

After seeing the strong statistical significant differences among treatment and control on increased well being, we chose to run another set of regressions with well being as our dependent variable as seen in figure 11 below.

Dependent variable:				
	Increased Well Being			
	(1)	(2)	(3)	(4)
Condition	0.631** (0.259)	0.530** (0.250)	0.612** (0.262)	0.550** (0.232)
Preferred.Gender_Male			-0.543** (0.262)	-0.400* (0.234)
Location_Manhattan			0.160 (0.291)	-0.057 (0.262)
Location_Queens			0.361 (0.491)	0.152 (0.438)
Ethnicity_Asian_Indian			-0.794 (1.350)	-0.540 (1.200)
Ethnicity_Black_American			-0.847 (0.997)	-0.071 (0.898)
Ethnicity_Hispanic			-0.872 (0.951)	-0.334 (0.851)
Class.Section_N			-0.406 (0.374)	-0.365 (0.332)
Class.Section_R			0.220 (0.361)	0.362 (0.321)
Class.Section_W			-0.157 (0.415)	0.027 (0.370)
letter_grade_B			-0.087 (0.533)	0.079 (0.474)
letter_grade_C			-0.115 (0.562)	0.049 (0.500)
letter_grade_D			-0.904 (0.767)	-0.916 (0.680)
b1.grad		0.234 (0.157)	0.285* (0.167)	0.225 (0.149)
b1.happ		0.307 (0.257)	0.247 (0.273)	0.088 (0.244)
effort				0.689*** (0.137)
Constant	3.540*** (1.240)	1.270 (0.817)	2.440* (1.400)	0.316 (1.310)
Observations	106	106	106	106
R2	0.055	0.134	0.236	0.405
Adjusted R2	0.046	0.109	0.108	0.298
Residual Std. Error	1.310 (df = 104)	1.260 (df = 102)	1.270 (df = 90)	1.120 (df = 89)
F Statistic	6.100** (df = 1; 104)	5.270*** (df = 3; 102)	1.850** (df = 15; 90)	3.790*** (df = 16; 89)
Note: *p<0.1; **p<0.05; ***p<0.01				

Figure 11: Stargazer ~ Regression of treatment and covariates on increased well being metric

From the results above, we find that treatment condition assignment does indeed have a positive significant impact on students' perception of well being. With no other controls, we find a coefficient of 0.631 which implies that being placed in the treatment group of gratitude journaling did indeed impact how students felt and improved their general sense of wellbeing. Adding in additional controls such as baseline gratitude and happiness scores did not reduce the

significance on condition though the coefficient decreased slightly. We also noted a more significant impact of subjective effort with wellbeing, further implying that the effort with which students conduct journaling regardless of topic may be the more impactful measure. Overall, we note that condition assignment remains statistically significant at the 5% level among all four models.

Conclusion

To conclude, our experiment investigating the impact of daily gratitude journaling on the happiness levels of Kate's high school freshmen students, revealed several significant insights. Our aim was to assess whether daily gratitude journaling could enhance happiness levels amongst the students; however, the results from our randomized controlled trial did not demonstrate significant differences in happiness or gratitude scores between the treatment and control groups at the conclusion of the intervention compared to the baseline. Contrary to our initial hypothesis, daily gratitude journaling did not lead to a notable increase in happiness levels among the treatment group compared to the control group, and there was no statistical time significance in scores throughout the duration of the experiment trial, as measured by the Oxford Happiness Questionnaire. The analysis section also highlighted that baseline happiness scores were consistently the most predictive of final happiness scores, suggesting individual stability in happiness traits over time. Additionally, despite the lack of significant differences in overall happiness and gratitude scores between groups, an intriguing finding emerged during the final survey collection: the treatment group reported significantly higher levels of well-being compared to the control group. This unexpected outcome implies that gratitude journaling may confer subjective benefits that were not captured in earlier rounds of questioning focused solely on happiness and gratitude metrics. Furthermore, while certain spurious correlations were observed with students' location (Manhattan) and ethnicities, additional investigation is required to interpret whether these findings represent meaningful associations or random occurrences. Ultimately, while our study did not reveal a significant impact of gratitude journaling on happiness levels among high school freshmen based on traditional metrics, the observed increase in well being scores among the treatment group suggests potential subjective benefits that merit further investigation. This highlights the importance of considering complex outcomes and subjective experiences in evaluating interventions aimed at promoting happiness and gratitude among high school students.

Limitations

There are various limitations we want to highlight in our experiment, as listed below.

- **Sample Characteristics:** Our sample involved a specific cohort of high school freshmen from a single school, Cristo Rey New York High School, with particular demographics (84% Hispanic, predominantly from Harlem/Manhattan and the Bronx). This

homogeneity limits the generalizability of the findings to broader populations, particularly those from different socioeconomic and cultural backgrounds.

- Attrition: A challenge arose due to attrition, with one student from the control group moving away during the study. This loss of a participant can impact the statistical power and integrity of the results, particularly in a relatively small sample size.
- Covariate Balancing: Despite efforts to balance covariates through randomization and blocking, some discrepancies were observed, particularly in ethnicity. This imbalance could introduce potential biases in the analysis and interpretation of results.
- Spurious Correlations: We identified spurious correlations with the location Manhattan, and ethnicity. However, these correlations may require further investigation to decipher whether they reflect meaningful associations or chance occurrences.
- Effect Size and Statistical Power: The experiment encountered limitations related to effect size and statistical power. Power calculations indicated that the observed effect sizes may not have been large enough to detect significant differences. This highlights the need for careful consideration of effect sizes in experimental design and interpretation.
- Subjective Measures: Our study focused on traditional metrics of happiness and gratitude but revealed unexpected findings related to subjective well being. This suggests the potential limitations of conventional measurement tools in capturing nuanced outcomes of gratitude interventions.
- Classroom Environment: Conducting the experiment within a classroom setting necessitated careful management of logistics, including privacy concerns, prompt distribution, and ensuring students' engagement and compliance with journaling activities.
- Interpretation of Findings: We faced challenges in interpreting non-significant results and unexpected outcomes, such as the observed increase in well-being among the treatment group. This underscores the complexity of evaluating subjective experiences and multifaceted outcomes in interventions targeting well being.

Future Work

There are several avenues for future work that we can take to to enhance the design of our experiment, these include:

- Refining Intervention Protocols: We can explore variations in gratitude journaling protocols such as in the frequency, duration, and add more varying content of prompts, to optimize the effectiveness of the intervention. We can also potentially experiment with

different kinds of formats and delivery methods tailored for the preferences of high school students.

- **Longer Term Follow Up:** Conducting a longer term follow up assessment past the immediate intervention period to evaluate the persistence and durability of any observed effects on happiness and gratitude, would provide valuable insights into the sustainability of gratitude journaling benefits over time.
- **Qualitative Inquiry:** Including quantitative measures with qualitative methods (e.g., interviews, focus groups) can help to capture variations of experiences and perceptions related to gratitude journaling. Qualitative data can illuminate subjective benefits and shed light on the underlying mechanisms driving changes in happiness levels.
- **Exploring External Factors:** Investigating potential factors in the high school students that may influence the impact of gratitude journaling on happiness scores, such as: personality traits, lifestyle, family life, etc. Understanding these factors could inform us further on the results of the intervention.
- **Spurious Correlation Analysis:** Further investigating spurious correlations identified in the study (e.g. location, ethnicity) to determine whether these associations have any meaningful implications or are rather chance occurrences. This would involve further analysis of the students' socio-cultural dynamics.
- **Addressing Attrition:** Developing strategies to mitigate attrition will enhance participants' engagement throughout the intervention period. Additionally, understanding factors contributing to attrition and participant adherence could optimize the intervention delivery.

Overall, by integrating these interventions in the future into our experiment, we can advance our understanding of the mechanisms underlying gratitude interventions and optimize their effectiveness in promoting happiness and gratitude in high school students. This iterative process will contribute to the development of strategies that support positive psychology in high schoolers.

Citations

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Appendix

Appendix 1: Journaling Prompts & Measurement Timeline

Treatment Prompts	Control Prompts	Day	Date	Measurements
Think about a personal challenge that you have experienced recently. How has this experience contributed to your personal growth and why you are thankful for the experience.	Create a SMART Goal you want to complete this week to get prepared for the exams. Remember: SMART = Specific, Measurable, Achievable, Relevant, and Time-Bound.	1	2/12/24 or 2/14/24	Baseline
Recall a recent moment when you felt particularly at peace or content. Where were you, and what were you doing? Reflect on why this moment was special and why you are thankful for it.	Rate your understanding on Photosynthesis from 1 (low) to 5 (high). Being honest explain to yourself why you think your understanding of Photosynthesis is that rating and what you need to do to improve your understanding of Photosynthesis.	2	2/14/24 or 2/15/24	
Consider the simple pleasures in life that bring you joy (a favorite food, a cozy moment, a favorite song). Choose one and describe why it's a source of gratitude for you.	What are some of the study strategies that you have utilized that have worked? What are some that hasn't? How can you improve on the strategies that did not work?	3	2/15/24 or 2/16/24	Mid treatment 2
Think about a person in your life that has had a positive impact on you. This could be a family member, a friend, a teacher, a coach, or a mentor. Write them a thank you letter explaining how they have helped, supported, or inspired you.	Did you achieve the SMART Goal you created for the week? If you went beyond the goal what did you do to help support yourself? If you did not reach your goal what were some of the blockers and how can you overcome them next week?	4	2/20/24	
Write about a time this week when someone did something unexpectedly kind for you, or when you did something kind for someone else. How did this act of kindness make you feel?	Reflect on a challenging topic you learned this semester. Describe what was difficult and what question/clarification you have about the topic.	5	2/21/24	
Take a moment to appreciate the beauty of nature around you. It could be the changing leaves, a clear sky, or a gentle breeze. Express gratitude for the wonders of the natural world.	Rate your understanding on Cellular Respiration from 1 (low) to 5 (high). Being honest explain to yourself why you think your understanding of Cellular Respiration is that rating and what you need to do to improve your understanding of Cellular Respiration.	6	2/22/24	Mid measurement 3
Trimester Exams and Spring Break 2/23/24 to 3/10/24				

Think about something you're looking forward to in the near future, and express gratitude for the anticipation and excitement it brings.	Create a SMART Goal you want to complete this week. Remember: SMART = Specific, Measurable, Achievable, Relevant, and Time-Bound.	7	3/11 or 3/12	
What does gratitude mean to you and what are some ways you express gratitude in your current day to day?	Identify a real-world application of a science concept you've studied, such as photosynthesis, cellular respiration, or the cell cycle. How does understanding this concept help explain the phenomenon? How can this knowledge be applied in practical or environmental solutions?	8	3/12 or 3/13	
Reflect on recent achievements and/or positive moments. What are you proud of achieving recently and why does it bring you a sense of gratitude?	Consider your future in science, whether it's related to your career aspirations or personal interest. Set a long-term SMART goal related to this vision. Outline the steps you plan to take in high school to work towards this goal.	9	3/13 or 3/14	
Consider the simple pleasures in life that bring you joy (a favorite food, a cozy moment, a favorite song). Choose one and describe why it's a source of gratitude for you.	Did you achieve the SMART Goal you created for the week? If you went beyond the goal what did you do to help support yourself? If you did not reach your goal what were some of the blockers and how can you overcome them next week?	10	3/14 or 3/15	Mid measurement 4
Think about a cultural or family tradition that means a lot to you. Describe the tradition and why it instills a sense of gratitude in your heart.	Reflect on a challenging topic you learned this unit. Describe what was difficult and what question/clarification do you still have about the topic.	11	3/18 or 3/19	
What are three things you could teach someone else about gratitude? Why have you chosen these things?	What is one topic in heredity that you would like to learn more about in class? Explain how you would like to learn about it (experiments, projects, movie).	12	3/19 or 3/20	
Acknowledge the growth you've seen in someone close to you. How does witnessing their growth inspire you and why are you grateful to be a part of their journey?	If you could study any topic in science what would it be? Explain why you want to learn about that topic. (EX: Computer science because you want to make video games one day)	13	3/20 or 3/21	
Think back on a favorite memory from your childhood. What makes this memory special and why are you grateful for it?	Rate your understanding on Heredity from 1 (low) to 5 (high). Being honest explain to yourself why you think your understanding of Heredity is that rating and what you need to do to improve your understanding.	14	3/21 or 3/22	Final Measurement 5

Appendix 2: Happiness Questionnaire and Additional Gratitude Statements based on The Oxford Happiness Questionnaire developed by psychologists Michael Argyle and Peter Hills at Oxford University

Self Check-in 4

B I U  

Below are a number of statements about happiness. Please indicate how much you agree or disagree with each by entering a number in the blank after each statement, according to the following scale:

- 1 = strongly disagree
- 2 = moderately disagree
- 3 = slightly disagree
- 4 = slightly agree
- 5 = moderately agree
- 6 = strongly agree

Please read the statements carefully, some of the questions are phrased positively and others negatively. Don't take too long over individual questions; there are no "right" or "wrong" answers (and no trick questions). The first answer that comes into your head is probably the right one for you. If you find some of the questions difficult, please give the answer that is true for you in general or for most of the time.

This form is automatically collecting emails from all respondents. [Change settings](#)

1. I don't feel particularly pleased with the way I am. *

1 2 3 4 5 6

strongly disagree ☐ ☐ ☐ ☐ ☐ ☐ strongly agree

Questions	Calculation Measurements
1. I don't feel particularly pleased with the way I am. (R)	Step 1. Items marked (R) should be scored in reverse: For example, if you gave yourself a "1," cross it out and change it to a "6." Change "2" to a "5" Change "3" to a "4" Change "4" to a "3" Change "5" to a "2" Change "6" to a "1".
2. I am intensely interested in other people.	
3. I feel that life is very rewarding.	
4. I have very warm feelings towards almost everyone.	
5. I rarely wake up feeling rested. (R)	
6. I am not particularly optimistic about the future. (R)	
7. I find most things amusing.	Step 2. For the Happiness Score: Add the numbers for all 1 to 29 questions. (Use the converted numbers for the 12 items that are reverse scored.)
8. I am always committed and involved.	Step 3. Divide step 2 by 29. So your happiness score = the total (from step 2) divided by 29.
9. Life is good.	
10. I do not think that the world is a good place. (R) _	Step 4. For the Gratitude Score: Add the numbers for all 3 gratitude questions
11. I laugh a lot.	Step 5. Divide step 4 by 3.
12. I am well satisfied about everything in my life.	
13. I don't think I look attractive. (R)	Step 6. For the Happiness and Gratitude Combined Score: Divide the total points (step 1 and step 4) and dived by 32.
14. There is a gap between what I would like to do and what I have done. (R)	
15. I am very happy.	The statements were converted into a Google form use the scale 1 = strongly disagree to 6 = strongly agree and distributed using student emails. Students completed the surveys during class after a 5-minute journaling time. Students used computers, laptops, tablets, and cell phones to access the Google form. If a student was absent on the day of the surveys they completed it the next day in class. Surveys were done on the last day of class of each week during the intervention.
16. I find beauty in some things.	
17. I always have a cheerful effect on others.	
18. I can fit in (find time for) everything I want to.	
19. I feel that I am not especially in control of my life. (R)	
20. I feel able to take anything on.	
21. I feel fully mentally alert.	
22. I often experience joy and elation.	
23. I don't find it easy to make decisions. (R)	
24. I don't have a particular sense of meaning and purpose in my life. (R)	
25. I feel I have a great deal of energy.	
26. I usually have a good influence on events.	
27. I don't have fun with other people. (R)	
28. I don't feel particularly healthy. (R)	
29. I don't have particularly happy memories of the past. (R)	
Additional Questions:	
30. I have a lot of people I am grateful for in my life.	
31. I feel gratitude.	
32. I find it easy to identify things I am grateful for.	

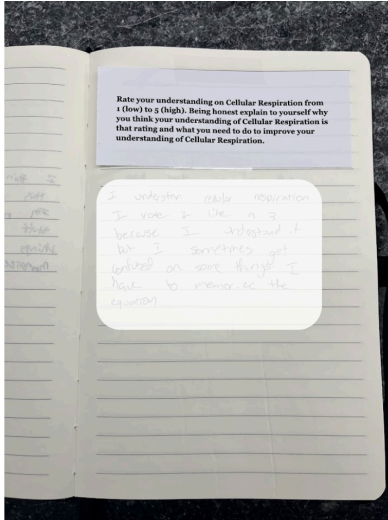
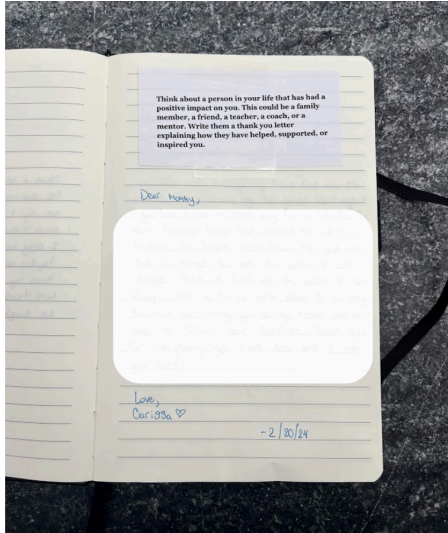
Appendix 3: Additional Final Questionnaire Statements and Student Responses Examples

Condition	Journaling has increased my overall sense of well-being. (out of 6)	Rank the amount of effort you applied to the journaling. (out of 5)	Explain your rank of effort, why was it important for you to add the effort you did to your journaling?	Reflecting on your experience with journaling, what aspect of journaling has been the most meaningful to you? Please give details.	Do you think that journaling has helped you appreciate the positive aspects of your life or increased your ability to cope with a difficult situations? Please explain or give an example.
Treatment	4	3	I just write it doesnt really feel any diff after	I just like writing.	It's given me a few things to think about
Treatment	5	4	It was important for me to add the effort I do to my journaling because it gives me a sense that I can do something and do it well.	When I write about the topics you give me. I really enjoy writing a lot so it's nice to know that I have a way to express it even if it's about a topic and not abstract.	Journaling has helped increase my ability to cope with difficult situations because it lets me write down how I feel about things and it helped me think of other things other than stress about certain things.
Treatment	4	5	TO let all o my thoughts out, and what feels like im thinking out loud	Who i feel gratitude to in my life	Yeah increasingly, by going back and really thinking what i am grateful for. But at the same time no because my problems are still going to be there, but i have to deal with it either way journal or not, no changes exsept my confidence.
Treatment	4	3	i just wrote whatever came to mind first since I know it would be what I would do.	I guess just the ones where I answered more on others.	Yes, because it helps me think about what I feel and helps me think about who I am.
Control	2	3	sometime I don't know what to write about the topic and sometimes it's not something I would like to journal about	I like journaling in my own time about personal problems and situations I use it as a coping mechanism	Not really mostly because there school related topics or questions
Control	3	4	I always try to answer the questions asked truthfully.	The time I have to just write my feelings.	I think journaling has increased the ability to cope with a difficult situations because I can always just write my feelings out during that time.
Treatment	4	3	Sometimes the prompts were fun so I put more effort into them than others	Writing in silence is sometimes peaceful and lets me have thinking time away from school	Sometimes because the prompts help me realize what I have in life and im grateful for
Control	5	3	Just because some things are are to hard to talk about	Being able to not talk about what we learn only	Yes,because it help you fix your mind when it's all over the place
Control	4	4	I added this effort because I though it would be useful to set a goal for myself.	The aspect of journaling that has been the most meaningful to me is with what I want to be able to study.	I think that journaling has a positive effect of me because that way I am able to set myself on a goal that I want to accomplish and write out what I am thinking of what I want to do.
Control	5	4	Cause i can see what i need and see what i KNEW i needed	The writing and the reflection i get. I get to somewhat better my handwriting and be able to reflect on how im feeling	Yes, i fee slightly more happy
Control	3	3	Not really	Not really	Not really
Treatment	3	3	It was important so our teacher can know	All of them i guess	A littlw
Control	3	3	no	no	no

In the final survey, five additional questions were added about student's journaling experiences and their view on its impact on their happiness and academic performance. With two questions on the scale of 1 (strongly disagree) to 6 (strongly agree) and three written response questions.

Appendix 4: Picture Documentation of Journaling

Below are two examples from the same section, one of a student in the control group and one of a student in the treatment group. Their last names and their writing has been obscured to protect their identity and to protect their privacy on their written journals.

Control Example	Treatment Example
 A photograph of an open notebook. The right page has a purple sticky note with the text: "Rate your understanding on Cellular Respiration from 1 (low) to 5 (high). Being honest explain to yourself why you think your understanding of Cellular Respiration is that rating and what you need to do to improve your understanding of Cellular Respiration." Below the sticky note, the student has handwritten: "I understand cellular respiration. It's more like a 3 because I understand it but I sometimes get confused on some things. I have to memorize the equation." The left page has some faint, illegible handwriting.	 A photograph of an open notebook. The right page has a purple sticky note with the text: "Think about a person in your life that has had a positive impact on you. This could be a family member, a friend, a teacher, a coach, or a mentor. Write them a thank you letter explaining how they have helped, supported, or inspired you." Below the sticky note, the student has handwritten a letter: "Dear Mommy, I'm thinking of you and how you've helped me so much. You've always been there for me and I love you. I'm thinking of you and how you've helped me so much. You've always been there for me and I love you. I'm thinking of you and how you've helped me so much. You've always been there for me and I love you." The letter is signed "Love, Cori 133" and dated "-2/20/24". The left page has some faint, illegible handwriting.